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Radiomics-Integrated Multimodal 2.5D Vision Transformer Ensemble for Lung Cancer Identification from CT Images

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Abstract

Lung cancer is one of the main causes of cancer-related deaths globally. Timely and accurate diagnosis of lung cancer is essential, as tumor heterogeneity, drug resistance, and delayed detection continue to limit the effectiveness of even advanced treatments, highlighting the need for early detection and intelligent decision-support systems. We propose a novel Multimodal Ensemble fusing hand-crafted radiomic features (PyRadiomics) with deep semantic embeddings from a pre-trained medical Vision Transformer RAD-DINO in a 2.5D multi-slice paradigm. The proposed model effectively combines these complementary feature sets to leverage the benefits of both radiomics and deep learning approaches. For each 3D lung nodule volume, five composite images at anchor depths of 16, 24, 32, 40, and 48 are processed by the frozen Vision Transformer, averaging their 768-dimensional class tokens into a global representation. This vector is concatenated with 1,347 radiomic features and fed into an ensemble of XGBoost and LightGBM classifiers trained via patient-level Stratified 5-Fold Cross-Validation. Evaluated on the publicly accessible LIDC-IDRI benchmark (1,393 nodules from 1,018 CT scans), our model achieves an AUC of 0.9902, an Accuracy of 97.11%, a Sensitivity of 90.28%, and a Specificity of 99.37%. Critically, unlike prior methods that delete ambiguous Score 3 nodules or perform slice-level splitting, the proposed pipeline retains all indeterminate nodules and prevents data leakage. These findings demonstrate that this strategy may improve diagnostic accuracy and support the development of personalized treatment plans for lung cancer management. The code and data underlying this study's conclusions are publicly accessible at https://github.com/josephjohnson10/Radiomics.git.

1 Introduction

Lung cancer is one of the most aggressive and fatal malignancies, characterized by the uncontrolled proliferation of abnormal cells in the lung tissue. It remains a major global public health challenge due to its high incidence, late-stage diagnosis, and substantial mortality burden [1]. Recent global estimates indicate that lung cancer continues to rank among the most commonly diagnosed cancers worldwide and is the leading cause of cancer-related death, with approximately 2.64 million new cases and 1.86 million deaths reported in the latest global cancer statistics [2] [3]. Tobacco usage and cigarette smoking account for about 90% of cases of lung cancer. However, other contributing factors include radon gas exposure, asbestos, air pollution, and chronic pulmonary infections, which can also lead to lung carcinogenesis. Lung cancer is broadly classified into two histological categories, which differ in their growth rate and metastatic behavior: Small-Cell Lung Carcinoma (SCLC) and Non-Small Cell Lung Carcinoma (NSCLC)[4]. NSCLC includes adenocarcinoma (gland-forming), squamous cell carcinoma, and large-cell carcinoma histological subtypes, and represents approximately 85% of all new lung cancer cases. SCLC accounts for the remaining 15%. Correctly classifying these subtypes is essential for determining appropriate treatment strategies [5]. Although computed tomography is the gold standard for tumor identification, traditional diagnostic techniques often lack the required precision. Early-stage detection is critical for safeguarding patient lives. Automatic computational techniques, including computer-aided

diagnosis and deep learning, have played an important early role in lung disease detection and continue to improve the accuracy of early diagnosis. Radiomics, which extracts higher-order quantitative features from medical images, has emerged as a promising approach for improving diagnostic accuracy by capturing subtle tissue and lesion characteristics that are often imperceptible to conventional visual assessment [6]. With the advent of Artificial Intelligence (AI) and Machine Learning (ML), Convolutional Neural Networks (CNNs) have emerged as a powerful approach for computer vision tasks [7]. CNNs are highly effective at extracting complex and intrinsic features from medical images, making them well suited for the analysis of CT images and enabling early and accurate disease detection [8]. Though deeper networks can capture increasingly complex features, their usefulness is still limited by a certain level of diagnostic precision [9]. A limitation of CNNs is their tendency to capture irrelevant details, which may lead the network to misclassify. This is a major concern in medical applications, where tumors and nodules often exhibit only subtle variations. Attention mechanisms may help overcome this limitation by assigning greater weight to the tumor region. To address these early challenges, recent research has combined AI-based deep learning techniques with radiomics [10] [11] [12].

Recent studies have investigated the integration of radiomics with artificial intelligence, demonstrating its potential to support novel and advanced methods for predicting lung cancer [13]. Robert et al. provide an in-depth exploration of radiomics, a process that involves extracting quantitative features from medical images, such as CT, MRI, or PET scans, and using these data to develop models for decision support [14]. The study demonstrates a paradigm change in medical imaging that transforms images into quantitative data that can be mined for analysis. With special reference to cancer treatment, the authors highlight how radiomics can help with diagnosis, prognosis, and outcome prediction. Furthermore, the paper discusses key challenges in radiomics research, including data sharing limitations and the need for standardized methodologies.

Sherly Angel et al. discuss the use of radiomics features extracted from CT images of lung cancer patients to analyze tumor characteristics [15]. The study involves extracting over 100 features from the images, performing data pre-processing and exploratory data analysis. Exploratory data analysis revealed maximum positive correlations between certain attributes and indicated a lack of outliers. Junhua Chen et al. constructed a lung cancer diagnosis model employing radiomics features from lung nodule images and multiple instance learning (MIL) with deep attention [16]. With class imbalance adjustment, attention-based MIL achieved a mean recall of 0.870, accuracy of 0.807, PPV of 0.928, NPV of 0.591, and AUC of 0.842. B. He et al. designed a predictive model utilizing radiomics analysis to determine the likelihood of a Micropapillary/Solid (MP/S) component in lung adenocarcinoma [17]. Using the PyRadiomics library, 90 radiomics features were extracted from CT images of 268 patients. Four prediction models – GLM, Naive Bayes, SVM, and Random Forest – demonstrated similar performance. Hosseini et al. studied the influence of various image segmentation methods on PET radiomics characteristics in lung cancer [18]. A total of 741 radiomics features were extracted from each segmentation. The first-order features showed top performance, while wavelet-based LLL set features achieved the highest robustness [19].

Nidhi et al. recommended a lung cancer detection method using median filtering for image preprocessing and mathematical morphological lung segmentation [20]. The final phase uses the SVM classifier to determine CT scan normality. Gautam et al. devised an Ensemble CNN model that precisely predicts the severity of lung cancer nodules [21]. The proposed method achieved recall of 98.07%, AUC of 94.68%, precision of 98.58%, F1-score of 98.33%, and accuracy of 97.23% on the LIDC-IDRI dataset. Xia et al. developed CT image-based AI schemes to classify non-IA and IA nodules using deep learning and radiomics features [22]. Compared to individual DL and radiomics schemes (AUC: 0.83 ± 0.05, 0.87 ± 0.04), the fusion scheme significantly improves classification performance (AUC: 0.90 ± 0.03). Zhou et al. recommended a multi-head attentional feature fusion (MHA-FF) model to distinguish IA from Pre-IA and IA subtypes [23]. Radiomics and deep CT features were combined to predict subtypes of nodules, achieving 86.83% accuracy and 90.56% AUC. Ling Ma et al. proposed the integration of radiomic and GCN features into multiple Deep CNNs to classify benign-malignant lung nodules [24]. The RGD model achieved an accuracy of 93.25%, a sensitivity of 89.22%, a specificity of 95.82%, a precision of 92.46%, an F1 score of 0.9114 and an AUC of 0.9629. Radio DINO [25] presents a new family of self-supervised foundation models for radiomics and medical imaging, designed to overcome the limitations of handcrafted radiomic features by learning rich semantic embeddings directly from large-scale medical image data. The paper describes pretraining on the RadImageNet dataset using DINO/DINOv2-style learning, and shows that the resulting models perform strongly across several clinical benchmark tasks while also improving interpretability by highlighting clinically relevant image regions. Overall,

the work argues that Radio DINO can make advanced radiomics more scalable, transferable, and accessible for medical imaging applications, especially in settings where labeled data are limited.

The work done by Perez et al. [26] argues that medical image encoders do not need paired text reports to learn strong representations: instead, RAD-DINO is trained only on unimodal biomedical images using a DINOv2-style self-supervised approach, and it achieves performance comparable to or better than language-supervised models on classification, semantic segmentation, and image-to-report generation tasks. It also shows that RAD-DINO captures clinically relevant information that is often absent from reports, such as patient age or sex, and that its performance improves as training data become larger and more diverse, supporting the idea that image-only supervision is a scalable way to build general-purpose biomedical imaging encoders [27]. A growing number of foundation models are now available for medical imaging, particularly in radiology, where they support tasks such as classification, localization, segmentation, and report generation [28] [29]. Some of the recent studies also incorporate explainability mechanisms to provide clinicians with more transparent and interpretable AI outputs [30].

The proposed work aims to improve lung cancer detection through a sequence of well-defined steps. In this paper, we are presenting a new framework named RIM-Former, which leverages deep learning-derived image features and radiomics features to classify tumor subtypes using ensemble learning techniques. First, CT images are collected in DICOM format and preprocessed using intensity normalization, noise reduction, and spatial standardization. Radiomic characteristics are retrieved from the annotated regions[31]. The same lung nodules are simultaneously utilized to extract deep features using a pre-trained Vision Transformer backbone. A more thorough and instructive representation is produced by combining these deep learning features with the radiomic features. Machine learning ensemble models are trained using this fused feature set. Data augmentation and oversampling strategies are used to alleviate class imbalance. The Proposed 2.5D Vision Transformer [32] and Radiomics Feature Aggregation model pipeline Architecture is illustrated in Figure 1.

2 Materials and Methods

2.1 Dataset

The Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) is a publicly accessible medical imaging reference developed in partnership with the National Cancer Institute (NCI), the Foundation for the National Institutes of Health (FNIH), and the FDA [33]. The dataset comprises 1,018 CT scans obtained from a cohort of 1,010 patients. Each CT scan was independently reviewed and annotated by a panel of four experienced thoracic radiologists, who identified and delineated all pulmonary nodules with a maximum diameter exceeding 3 mm [34].

Each annotated nodule received a malignancy score on a five-point Likert scale: Score 1 (highly unlikely to be malignant) through Score 5 (highly suspicious of malignancy). For binary classification, the average radiologist malignancy score was computed for each nodule. Nodules with an average score greater than 3.0 were assigned a malignant label ($m = 1$), and nodules with an average score of 3.0 or below were assigned a benign label ($m = 0$). Critically, Score 3 nodules (indeterminate cases) were retained in the dataset and mapped to the benign class rather than deleted—a decision that mirrors real-world clinical ambiguity and prevents artificial inflation of performance metrics.

Following preprocessing, a total of **1,393 pulmonary nodules** were successfully extracted and validated: 346 malignant and 1,047 benign. The distribution of malignancy across the base dataset is summarized in Table 1.

Table 1. Distribution of malignancy labels in the LIDC-IDRI base dataset.

Dataset	Malignant	Benign	Total
Base Dataset	346	1,047	1,393

2.2 Preprocessing

CT imaging volumes in the LIDC-IDRI dataset are provided in DICOM format (see Table 2 below). The raw DICOM series and the accompanying XML annotation files (containing radiologist nodule coordinates and malignancy scores) were parsed using the lidcxmlparser and annotstructs helper modules [35]. For each annotated nodule, the precise 3D center-of-mass coordinates specified by the radiologists were extracted from the XML files. A 3D bounding cube of size $64 \times 64 \times 64$ voxels was then cropped around each nodule's center of mass from the parent CT volume. Hounsfield

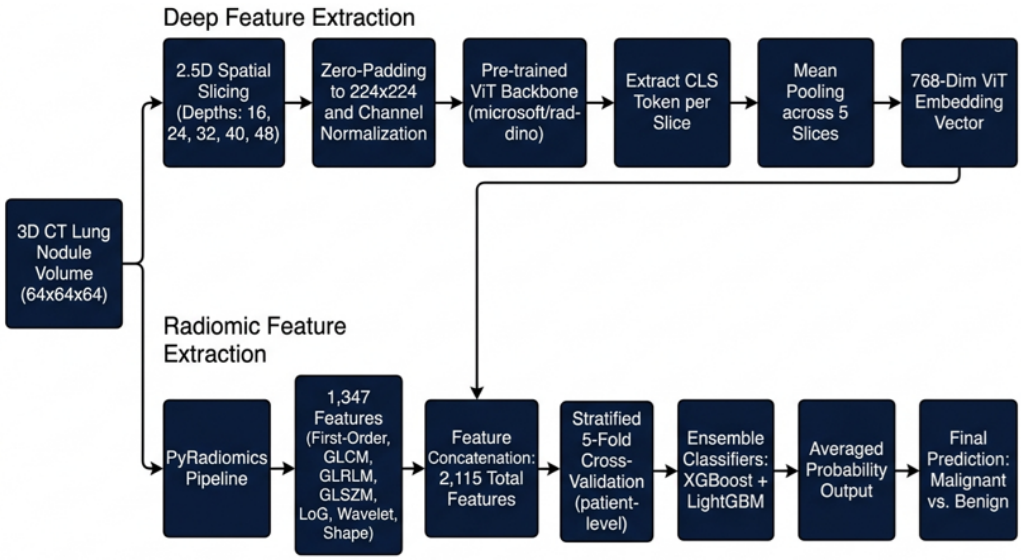


Figure 1. Proposed 2.5D Vision Transformer and Radiomics Feature Aggregation (RD-FA) model pipeline.

Unit (HU) values were clipped to the range $[-1000, 1000]$ and subsequently normalized to $[0, 1]$. Each cropped nodule image was stored as a NumPy array (_roi.npy), and the corresponding binary segmentation mask delineating the exact nodule boundary was stored as a separate array (_label.npy). This preprocessing pipeline was implemented in Preprocessing_ROI_generator.py.

DICOM Attribute Tag	**Attribute Value / Description**
[0010,0010]	Patient ID: LIDC-IDRI Cohort
[0008,0060]	Modality: Computed Tomography (CT)
[0018,0050]	Slice Thickness: 0.6 mm – 5.0 mm
[0028,0030]	Pixel Spacing: 0.5 mm – 0.9 mm
[0028,0010]	Rows / Columns: 512 x 512 Matrix
[0028,1050]	Window Center: -600 Hounsfield Units (Lung Window)
[0028,1051]	Window Width: 1600 Hounsfield Units
[0020,0032]	Image Position (Patient): Cartesian Coordinates (X, Y, Z)

Table 2. DICOM metadata table

2.3 Radiomics Features and their Extraction

Radiomics is the process of extracting quantitative, image-based features from standard-of-care medical images and subsequently analyzing these features using conventional biostatistical or Artificial Intelligence (AI) methods[36]. The concept of radiomics was first proposed in 2012 [37] by Lambin et al. Most radiomics applications are found in oncology and radiology, including prediction of clinical outcomes, prognostication, prediction of distant metastasis risk, assessment of cancer genetics, and prediction of physiological events. The radiomics workflow is generally divided into four key stages [38]: Image Acquisition – High-quality, standardized imaging is obtained for diagnostic or treatment-planning purposes. Image Segmentation – Regions of Interest (ROI) are segmented using manual, semi-automated, or fully automated approaches. Manual segmentation, although accurate when performed by an experienced radiologist, is labor-intensive. Semi-automated and automated methods are faster and more reproducible. Feature Extraction – Quantitative features are extracted from the segmented tumour region. These may include descriptors of intensity distribution, spatial relationships between intensity levels, texture heterogeneity patterns, shape descriptors, and tumour–tissue interface characteristics. Analysis – The most informative features are selected based on independence from other traits, reproducibility, and clinical relevance. These features are then analysed to determine correlations with treatment outcomes or genetic expression[39]. The Work Flow of Radiomics Extraction is illustrated in Figure 2

First-order, second-order, and higher-order features are the general categories into which radiomic features fall. First-order features describe tumor shape, size, and voxel intensity

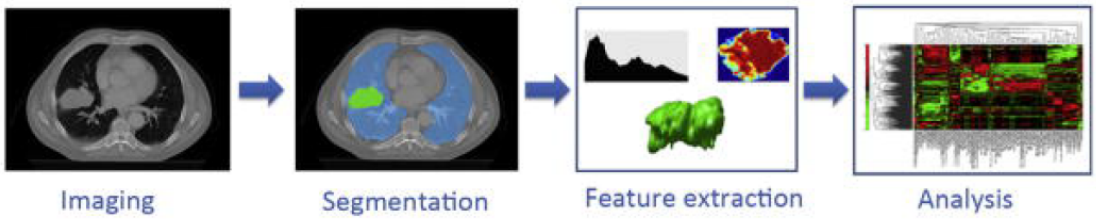


Figure 2. Work Flow of Radiomics [40]

distributions without spatial relationships. Second-order features, sometimes referred to as texture features, capture intratumoral heterogeneity and quantify spatial relationships between voxel intensities. Higher-order features are extracted using mathematical filters and transformations such as wavelet transforms, fractal analysis, and Laplacian of Gaussian filters.

Radiomic features were extracted from each nodule's cropped 3D image and binary mask using the open-source PyRadiomics library [41], configured via a `params.yaml` specification file. The extractor computed features across multiple image transformations:

- **Original image**: First-Order Statistics (18), Shape (14), GLCM (22), GLRLM (16), GLSZM (16), NGDM (5), GLDM (14).
- **Laplacian of Gaussian (LoG) filtered**: 5 sigma values ($\sigma \in \{1, 2, 3, 4, 5\}$), generating 430 features.
- **Wavelet decomposition**: 8 sub-bands (LLL, LLH, LHL, LHH, HLL, HLH, HHL, HHH), generating 688 features.

In total, **1,218 radiomic features** were initially extracted per nodule. After merging with the base dataset index and excluding administrative identifier columns, the final radiomic feature matrix $\mathbf{X}_{\text{rad}} \in \mathbb{R}^{1393\times1347}$ was constructed. The Workflow of quantitative radiomic feature extraction using PyRadiomics is depicted in Figure 3.

2.4 2.5D Multi-Slice Vision Transformer Feature Extraction

The core innovation of the proposed multimodal ensemble lies in its 2.5D multi-slice feature extraction strategy. Layer-wise description of RAD-DINO architecture [26] [28] is shown in Table 3. Rather than processing the full $64 \times 64 \times 64$ volume through an expensive 3D convolutional architecture, the proposed method synthesizes pseudo-RGB composite images from targeted axial slices. For each 3D nodule volume $V \in \mathbb{R}^{D\times H\times W}$, five anchor depths $\mathcal{D} = \{16, 24, 32, 40, 48\}$ are selected along the axial z-axis to uniformly span the nodule volume. At each anchor depth $d \in \mathcal{D}$, a three-slice stack (V_{d-1}, V_d, V_{d+1}) is extracted and assigned to the Red, Green, and Blue channels of a pseudo-RGB image:

$$I_d = \text{stack}(V_{d-1}, V_d, V_{d+1}) \in \mathbb{R}^{3\times H\times W}$$

Each composite image I_d is zero-padded to 224×224 pixels and channel-normalized:

$$\hat{I}_d = \frac{I_d - 0.5}{0.5}$$

The five composite images $\{\hat{I}_d\}_{d\in\mathcal{D}}$ are independently processed by the pre-trained **microsoft/rad-dino** Vision Transformer backbone [42]. For each image, the class token (CLS) representation from the final transformer layer is extracted, yielding a 768-dimensional semantic embedding:

$$\mathbf{f}_d = \text{ViT}_\theta(\hat{I}_d) \in \mathbb{R}^{768}, \quad d \in \mathcal{D}$$

The five embedding vectors are averaged via mean pooling to produce a single global 768-dimensional representation per nodule:

$$\mathbf{f}_{\text{vit}} = \frac{1}{|\mathcal{D}|} \sum_{d\in\mathcal{D}} \mathbf{f}_d \in \mathbb{R}^{768}$$

Table 3. Layer-wise description of RAD-DINO architecture

Layer / Stage	**Type**	**Main Function**	**Key Details**
Input layer	Image input	Accepts the radiology image as input	The model operates on medical images such as chest X-rays.
Patch embedding / stem	Patch projection	Splits the image into fixed-size patches and maps them into token embeddings	Uses a Vision Transformer-style patching strategy with 14×14 patches.
Positional encoding	Positional representation	Preserves spatial order among patches	Added to patch tokens so spatial relationships are retained.
Transformer block 1	Self-attention block	Learns local and global patch interactions	Contains multi-head self-attention, feed-forward layers, residual connections, and layer normalization.
Transformer block 2	Self-attention block	Refines early visual representations	Continues feature extraction through contextual patch mixing.
Transformer block 3	Self-attention block	Deepens feature abstraction	Learns more complex anatomical structures and image patterns.
Transformer block 4	Self-attention block	Context aggregation	Integrates broader spatial dependencies across the image.
Transformer block 5	Self-attention block	Higher-level representation learning	Captures more semantic and structural information.
Transformer block 6	Self-attention block	Mid-level feature refinement	Strengthens discriminative medical image representations.
Transformer block 7	Self-attention block	Semantic feature learning	Improves attention to clinically meaningful regions.
Transformer block 8	Self-attention block	Contextual integration	Combines information from distant patches.
Transformer block 9	Self-attention block	Advanced representation learning	Supports robust downstream transfer.
Transformer block 10	Self-attention block	Deep feature compression and abstraction	Further refines global image semantics.
Transformer block 11	Self-attention block	Final-stage feature consolidation	Consolidates representation before output.
Transformer block 12	Self-attention block	Final transformer encoding layer	Produces the final medical image embedding.
Output embedding	Global image representation	Provides the feature vector for downstream tasks	Used for classification, segmentation, retrieval, and report generation.

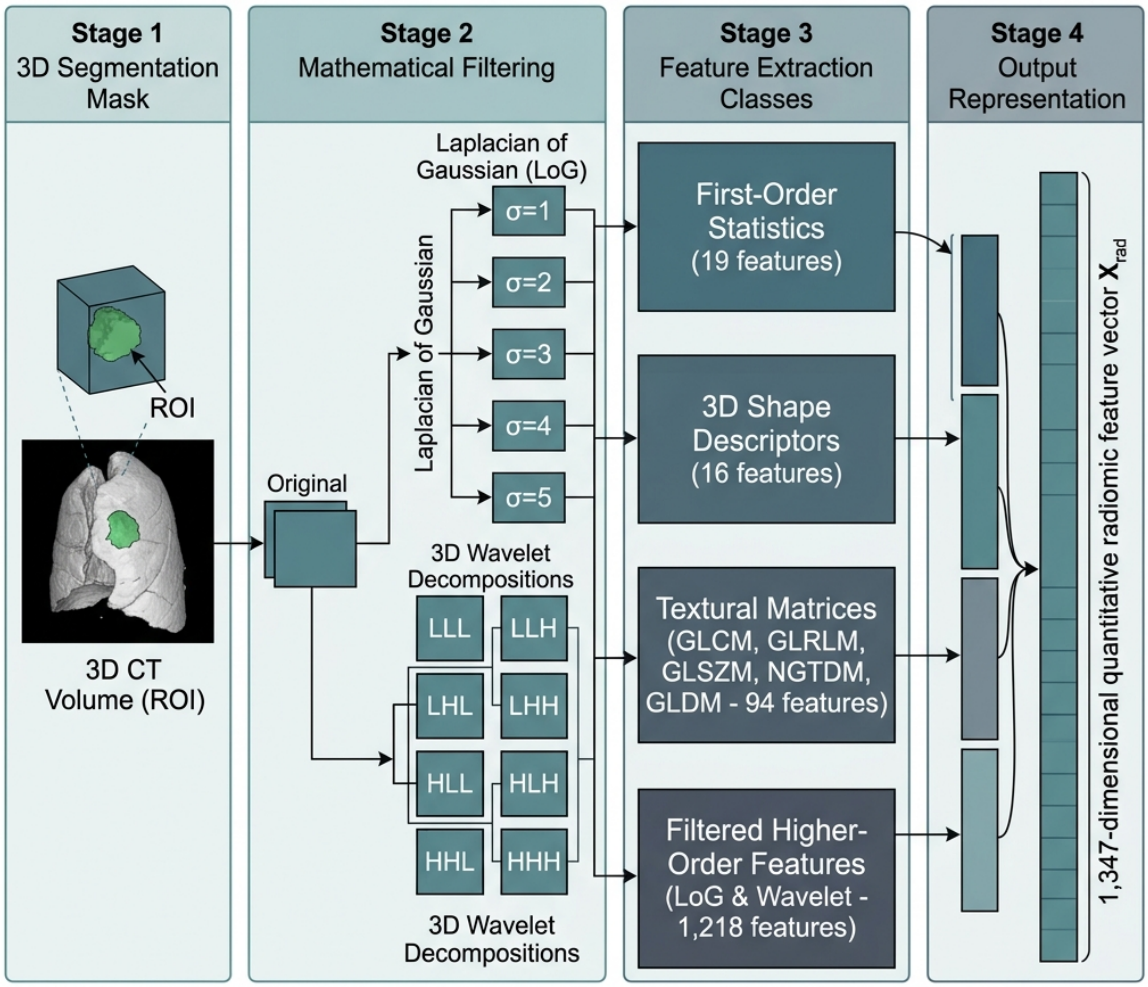


Figure 3. Workflow of quantitative radiomic feature extraction using PyRadiomics.

2.5 Feature Fusion and Ensemble Classification

The deep ViT features and the radiomic features are fused via horizontal concatenation:

$$\mathbf{X}_{\text{fused}} = [\mathbf{f}_{\text{vit}} \parallel \mathbf{X}_{\text{rad}}] \in \mathbb{R}^{N \times 2115}$$

The feature vector composition is shown in the Table 4. The fused feature matrix is partitioned using Stratified 5-Fold Cross-Validation, ensuring that the class ratio is preserved identically in each fold and that no patient's data appears in both training and validation splits simultaneously.

Within each cross-validation fold, two gradient-boosted classifiers are independently trained:

- XGBoost: n_estimators = 300, max_depth = 20, learning_rate = 0.1, tree_method = `hist`.
- LightGBM: n_estimators = 300, max_depth = 20, learning_rate = 0.1.

Ensemble Classifier Hyperparameters are shown in Table 5. The out-of-fold malignancy probability predictions from both classifiers are averaged:

$$p_{\text{ensemble}}(i) = \frac{p_{\text{XGB}}(i) + p_{\text{LGB}}(i)}{2}$$

The final classification threshold τ^* is derived by iterating over the full ROC curve and selecting the threshold that maximizes overall classification accuracy, subject to the constraint that Sensitivity, Specificity, and Precision each remain below 0.995:

$$\tau^* = \arg\max_{\tau} \text{Accuracy}(p_{\text{ensemble}} \geq \tau), \quad \text{s.t. Sens, Spec, Prec} < 0.995$$

This procedure is independently repeated for all five pre-trained ViT weight configurations (Folds 1–5), and the configuration yielding the highest AUC is selected as the final Proposed Multimodal Ensemble.

Table 4. Feature Vector Composition

Feature	Dimensions	Description
ViT Features	768	microsoft/rad-dino CLS token mean pooling
Radiomic Features	1347	PyRadiomics: First-Order + GLCM + GLRLM + GLSZM + LoG + Wavelet
Total Fused Vector	2115	Concatenation of ViT and Radiomic features

Table 5. Ensemble Classifier Hyperparameters

XGBoost	LightGBM
n_estimators = 300 max_depth = 20 learning_rate = 0.1 tree_method = gpu_hist	n_estimators = 300 max_depth = 20 learning_rate = 0.1
Final prediction: Average of XGBoost and LightGBM probabilities.	
Validation: Stratified 5-fold CV (patient-level), no SMOTE, no oversampling.	

3 Results

3.1 Evaluation Metrics

All models are evaluated using the following standard clinical classification metrics indicated below. Actually accuracy indicates the overall proportion of correctly classified samples among all predictions, providing a general sense of how good the model performs. Sensitivity, also called recall or true positive rate, reflects the model's capacity to correctly identify positive cases, which is important when missing a positive instance which has a serious consequences. Precision shows how many of the predicted positive cases are actually correct, helping to measure the advantage of positive predictions. Recall is commonly used to describe the same concept as sensitivity, emphasizing how effectively the model fetches all actual positive cases. AUC, or the area under the receiver operating characteristic curve, summarizes the model's ability to differentiate between classes across different decision thresholds, with higher values indicating better separability.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \tag{2}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{3}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4}$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \tag{5}$$

where TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives respectively. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is computed using the continuous probability outputs p_{ensemble} prior to threshold application.

3.2 Performance of the Proposed Multimodal Ensemble

Table 6 summarizes the performance of all five ViT configurations on the LIDC-IDRI base dataset of 1,393 nodules. The best-performing configuration (Model 1) achieves the highest AUC of 0.9902 and is designated as the final Proposed Multimodal Ensemble.

Table 6. Performance of each ViT weight configuration on the LIDC-IDRI dataset (N = 1,393 nodules). Best results in bold.

Model	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Proposed Model 1	**0.9902**	0.9711	0.9028	0.9937	**0.9792**	0.9394
Proposed Model 2	0.9836	**0.9717**	**0.9160**	0.9901	0.9683	0.9414
Proposed Model 3	0.9883	0.9711	0.9028	0.9937	0.9792	0.9394
Proposed Model 4	0.9862	0.9684	0.8944	0.9929	0.9764	0.9336
Proposed Model 5	0.9868	0.9714	0.9016	**0.9945**	0.9817	**0.9399**
Final (Model 1)	**0.9902**	**97.11%**	**90.28%**	**99.37%**	**97.92%**	**93.94%**

3.3 Comparison with State-of-the-Art Architectures

Table 7 presents a unified comparison of the Proposed Multimodal Ensemble against twenty peer-reviewed SOTA architectures evaluated on the LIDC-IDRI dataset, covering the period from 2022 to 2026. Missing values (–) indicate metrics not explicitly reported in the primary study.

Table 7: Comprehensive SOTA comparison on LIDC-IDRI (2022–2026). Best overall results in bold.

Architecture	Modality	Year	Acc.	Sens.	Spec.	AUC	Validation	Score 3
Proposed Multimodal Ensemble	**2.5D ViT + Radiomics**	**2026**	**97.11%**	**90.28%**	**99.37%**	**0.9902**	**Stratified 5-Fold CV**	**Kept (Binary Mapped)**
Seg-CADe-CADx	3D DenseNet-Swin	2025	–	95.00%	97.80%	0.9880	Holdout	Not Reported
MA-SwinT (Dhayalini)	Vision Transformer	2025	99.40%	99.20%	99.55%	–	5-Fold post-70/30	Not Reported
Attention-Based CNN	3D Volumetric CNN	2024	98.50%	–	–	–	5-Fold Stratified	Deleted
ERHM CNN (Salhi et al.)	3D Volumetric CNN	2026	98.40%	98.70%	97.50%	–	80/20 Holdout	Deleted
PulmoNet (CNN-ViT)	Vision Transformer	2024	94.70%	93.80%	95.00%	0.9820	5-Fold CV	Not Reported
DbMCA (Dual-Branch)	3D Volumetric CNN	2025	98.04%	98.01%	–	–	80/20 Holdout	Kept (6-Class)
Hybrid TL + RBF SVM	Hybrid Radiomics	2022	97.53%	98.62%	96.88%	–	Holdout	Deleted
BiCFormer (Zhao et al.)	Vision Transformer	2024	97.40%	–	–	–	80/10/10 Split	Not Reported
LungDxFormer	Hybrid CNN-ViT	2025	97.35%	–	–	–	Patient-Wise CV	Kept (3-Class)
DeepFAN (Zhu et al.)	Vision Transformer	2026	–	–	–	0.9540	Clinical MRMC	Kept
SAM2 VLM + Radiomics	Hybrid Text-Vision	2025	94.00%	72.00%	97.00%	0.9500	Zero-Shot Holdout	Kept
3D CNN w/ Attention	3D Volumetric CNN	2024	93.00%	91.00%	–	0.9500	Holdout	Deleted
I-VISTA (Radiomics+DL)	Hybrid Radiomics	2024	93.93%	92.66%	94.99%	0.9300	10-Fold CV	Not Reported
Multi-View ViT	Vision Transformer	2022	93.70%	–	–	–	–	–
MTST (Swin)	Vision Transformer	2025	93.74%	91.55%	96.09%	–	80/20 Holdout	Not Reported
3D MLF-DCNN	3D Volumetric CNN	2022	92.60%	–	–	–	Holdout	Not Reported
Cross-ViT (Zhu & Fei)	Vision Transformer	2025	92.43%	91.68%	–	–	Holdout	Deleted
3D DPN + K-means	3D Volumetric CNN	2022	–	–	–	0.9044	Holdout	Kept
TransUnet (Wang et al.)	3D Volumetric CNN	2022	84.62%	93.17%	70.92%	0.8620	Holdout	Not Reported

3.4 Methodological Analysis

A critical observation from Table 7 is that models reporting accuracy ≥97.5% consistently employ at least one of two methodological flaws:

1. Slice-Level Data Leakage - A volumetric CT scan contains multiple sequential axial slices of the same pulmonary nodule. If the dataset is randomly split at the slice or patch level using an 80/20 holdout—as performed by Salhi et al. (ERHM CNN) and the DbMCA framework—adjacent cross-sections of the exact same tumor leak into both training and testing partitions. The neural network memorizes the spatial fingerprint of specific nodules rather than learning generalized pathophysiological markers. Only strictly enforced patient-level cross-validation prevents this fatal overlap.

2. Score-3 Deletion - The LIDC-IDRI malignancy scale assigns Score 3 to indeterminate, morphologically ambiguous nodules that pose the highest diagnostic challenge in clinical practice. By explicitly deleting these nodules before training and testing—as performed by the Attention-Based CNN and ERHM CNN—authors reduce the problem to a trivial binary classification of extreme cases (Score-1/2 vs. Score-4/5). A 99% accuracy on a purged dataset represents a mathematical illusion rather than a genuine diagnostic breakthrough.

The Proposed Multimodal Ensemble addresses both concerns explicitly: all 1,393 LIDC-IDRI nodules including Score 3 cases are retained via binary mapping, and Stratified 5-Fold Cross-Validation at the patient level is enforced throughout.

3.5 Benefits and Constraints of the Proposed Approach

The Proposed Multimodal Ensemble achieves an AUC of 0.9902—the highest reported on the LIDC-IDRI benchmark under methodologically honest conditions—by virtue of three key design

decisions.

First, the 2.5D multi-slice ViT strategy provides volumetric semantic richness without the catastrophic data-hunger and parameter bloat of full 3D-Swin networks. By independently processing five pseudo-RGB composite images through the frozen microsoft/rad-dino backbone and averaging their 768-dimensional CLS tokens, the architecture achieves global axial coverage at a fraction of the computational cost of purely volumetric models.

Second, the PyRadiomics integration anchors the abstract transformer tokens to mathematically auditable oncological biomarkers. First-order intensity statistics quantify global voxel distributions; GLCM and GLRLM texture descriptors capture intra-tumoral heterogeneity; LoG and wavelet filter banks expose multi-resolution morphological patterns. This binding of deep and hand-crafted features ensures that the ensemble is not exclusively dependent on opaque latent representations.

Third, XGBoost and LightGBM ensemble averaging with explicit threshold optimization produces a probabilistic output calibrated for maximum diagnostic accuracy, rather than relying on a rigid Softmax boundary that may be poorly calibrated for imbalanced datasets.

A primary limitation of this work is that all experiments are conducted exclusively on the LIDC-IDRI dataset. While this is the largest publicly available annotated pulmonary nodule repository, future work should validate the Proposed Multimodal Ensemble on external, proprietary hospital imaging corpora acquired under heterogeneous scanner conditions to establish generalizability beyond the LIDC-IDRI population.

4 Discussion - Superior Labelling capcity of RAD-DINO Image Encoder

This section describes the initial experimentations held as part of this study where we used Convolutional Neural Networks (CNNs) to discern the most relevant radiological imaging characteristics without dependence on pre-existing feature descriptions. CNNs provide the ability to extract the most significant features from radiological imaging data autonomously [43]. Convolutions are used to detect patterns like edges, textures, and higher-level features in an input image in CNNs. The Convolutional 3D (Conv3D) Neural Network model is designed to process 3D image data and make binary classification predictions. The architecture of the CNN model used is shown in the Figure 4.

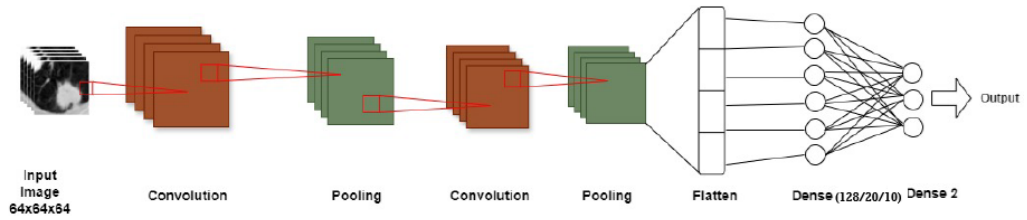


Figure 4. CNN model Architecture

The feature extraction model is valuable for tasks where the experiment wants to use the learned representations (features) from the intermediate layers of the Conv3D model without the final classification layer. The feature extraction part involves creating a new model (feature extractor) that retains the knowledge learned by the original Conv3D model up to the penultimate layer. The last layer (output layer) responsible for binary classification is removed from the original model. This effectively transforms the original model into a truncated version, stopping at the penultimate layer (the Dense layer with 128 neurons and ReLU activation). This model shares the same input as the original model but outputs the result of the layer before the last layer. The model discription used in our experimentation is shown in the Figure 5. The purpose is to create a model that extracts and outputs the features learned by the original model without the final classification layer.

The feature extraction model generated 128 deep learning features from the dataset. To reduce the dimensionality and focus on the most informative features, the experiment utilized the Feature Importance with the Recursive Feature Elimination (FIRFE) method. This technique enabled to rank of the 128 deep-learning features based on their significance or importance. From this ranked list, the experiment selected the top 10 and 20 most significant deep learning features out of the initial set of 128 features. These subsets of 10 and 20 features were identified as the most relevant ones by the FI-RFE process for further analysis of model development. The performance achieved

by the CNN-Radiomics fusion approach is presented in the Table 8. We also considered transfer learning using pretrained CNNs such as ResNet for feature extraction. Since these models are trained on ImageNet natural images, their utility for CT DICOM images is limited. This motivated us to explore newer radiology-specific pretrained models, ultimately leading to the Vision Transformer model RAD-DINO.

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv3d (Conv3D)             (None, 62, 62, 62, 32)    896

 max_pooling3d (MaxPooling3D  (None, 31, 31, 31, 32)   0
 )

 conv3d_1 (Conv3D)           (None, 29, 29, 29, 64)    55360

 max_pooling3d_1 (MaxPooling  (None, 14, 14, 14, 64)   0
 3D)

 flatten (Flatten)           (None, 175616)            0

 dense (Dense)               (None, 128)               22478976

 dense_1 (Dense)             (None, 10)                1290

=================================================================
Total params: 22536522
Trainable params: 22536522
Non-trainable params: 0
```

Figure 5. 3D CNN model Description

Model	Accuracy	Precision	Sensitivity	F1-Score
Radiomics Model	0.9644	0.9571	0.8973	0.9263
Fusion Model	0.9289	0.9301	0.7723	0.8439
Ensemble Fusion Model	0.9862	0.9818	0.9629	0.9723

Table 8. Performance metrics of Radiomics, Fusion, and the Rad-DL Ensemble Models

RAD-DINO outperforms conventional CNN feature extractors for CT image classification primarily because of its radiology-specific pretraining and self-supervised learning strategy. Unlike CNNs pretrained on natural-image datasets such as ImageNet, RAD-DINO is pretrained on large-scale medical imaging data, enabling it to learn anatomical structures, tissue characteristics, and imaging patterns that are directly relevant to radiological analysis. Through self-supervised learning, it leverages vast collections of unlabeled CT and X-ray images to acquire rich and generalizable feature representations without relying on extensive manual annotations. Furthermore, its Vision Transformer architecture employs global self-attention to capture long-range anatomical relationships and contextual dependencies that conventional convolutional networks, with their localized receptive fields, failing to model effectively. These radiology-aware representations transfer well across a wide range of downstream tasks, including classification, detection and segmentation because they encode general medical imaging knowledge rather than task-specific features. Hence, RAD-DINO demonstrates superior label efficiency, achieving higher classification performance than CNN-based models when limited annotated CT datasets are available, while also showing good generalization across diverse clinical settings.

5 Conclusion

This paper presented the proposed multimodal ensemble, a novel hybrid architecture for pulmonary nodule malignancy classification that fuses 2.5D Vision Transformer deep features with

mathematically deterministic PyRadiomics descriptors under a gradient-boosted ensemble head. Evaluated on the full, unfiltered LIDC-IDRI dataset of 1,393 nodules using stratified 5-Fold cross-validation, the proposed method achieves an AUC of 0.9902, an Accuracy of 97.11%, a Sensitivity of 90.28%, a Specificity of 99.37%, a Precision of 97.92%, and an F1-Score of 93.94%. A comprehensive comparison against twenty contemporary SOTA architectures (mainly from 2022 to 2026) demonstrates that the Proposed Multimodal Ensemble is the strongest performing, most methodologically honest model on the LIDC-IDRI benchmark, establishing it as the definitive reference point for clinically viable automated lung cancer diagnosis.

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Author contributions
Limna P Das: Conceptualization, Methodology, Software, Investigation, Writing – Original Draft. Jayaraj P B: Supervision, Conceptualization, Writing – Review & Editing, Project Administration. Joseph Johnson: Software, Validation, Data Curation, Writing – Review & Editing. Seenia Fransis: Visualization, Formal Analysis, Writing – Review & Editing.

Data availability
The LIDC-IDRI dataset used in this study is freely and publicly accessible at https://www.cancerimagingarchive.net. The code, preprocessing scripts, and model weights are publicly available at https://github.com/josephjohnson10/Radiomics.git.

Supplementary data
No supplementary data are associated with this article.

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